

Trial Higher School Certificate Examination – Extension I Mathematics

Question One (12 marks)

Marks

- a) Find the perpendicular distance from the point $(1, 4)$ to the line $4y = 3x - 2$ **2**
- b) Fully factorise $2x^3 - 128$ **2**
- c) Differentiate with respect to x
- i) $y = \sin 2x$ **1** ii) $y = \log_e \sqrt{\frac{2x-1}{3x+2}}$ **2**
- d) Find the remainder when the polynomial $P(x) = x^4 - 2x^3 - 3$ is divided by $(x-2)$. **2**
- e) Evaluate $\int_{\sqrt{2}}^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$ **3**

Question 2 (12 marks) Start on a new page

- a) Find the acute angle between the lines $2x + y = 17$ and $3x - y = 3$ **2**
- b) Evaluate $\int_0^3 \frac{y}{\sqrt{y+1}} dy$, using the substitution $y = u^2 - 1$ **4**
- c) Prove that $\frac{\cos 2\theta}{\cos \theta - \sin \theta} = \cos \theta + \sin \theta$ **3**
- d) Find all solutions to: $\frac{1}{x(2-x)} \leq 0$ **3**

Question 3 (12 marks)**Marks**

- a) P (−7, 3), Q (9, 15) and B (14, 0) are three points and A divides the interval PQ in the ratio 3:1.

Prove that PQ is perpendicular to AB.

3

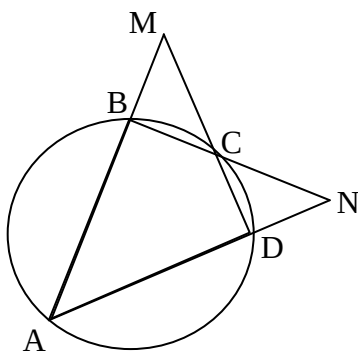
- b) If $f(x) = 2\sin^{-1}(3x)$ find the domain and range of $f(x)$

2

- c) Find the value of the constant term in the expansion of $\left(3x + \frac{2}{\sqrt{x}}\right)^6$

2

d)



In the figure ABM , DCM , BCN and ADN are straight lines and angle $AMD = \text{angle } BNA$.

- i) Copy the diagram onto the answer sheet and prove that angle $ABC = \text{angle } ADC$.
- ii) Hence, or otherwise prove that AC is a diameter.

3**2****Question 4 (12 marks)**

- a) Given $f(x) = \log_e(e^x + 1)$

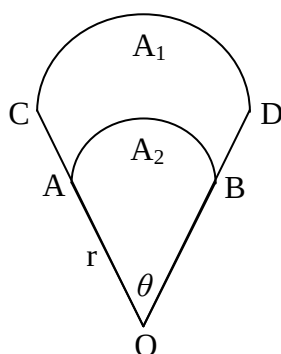
- i) Find the area bounded between $y = f(x)$, the x axis, $x = 1$ and $x = 5$ by the use of five ordinate values of Simpson's rule. Leave your answer correct to 2 decimal places.

3

- ii) By considering the second derivative of $f(x)$, state whether the application of the trapezoidal rule would over or under estimate the actual area described in part (i) above. Justify your answer.

3

b)



The diagram shows sector OCD which is formed by a piece of wire and a concentric arc AB , which is formed by a second piece of wire. The length of OA is r .

- i) Show that, if the area A_1 of region $ABDC$, is three times the area A_2 of region OAB , then $AC = r$. 2
- ii) If the total length of the two pieces of wire forming the figure is 24 cm, show that $\theta = \frac{24 - 4r}{3r}$. 2
- iii) Hence find the value of r so that the area of the sector OCD is a maximum. 2

Question 5 (12 marks)

- a) Solve for $0 \leq \theta \leq \pi$, $\cos \theta + 3 \sin \frac{\theta}{2} - 2 = 0$ 3
- b)
 - i) Sketch the curve $y = 4 \tan^{-1}(x)$ clearly showing its range. 2
 - ii) Find the volume of the solid formed when the area bounded by the curve $y = 4 \tan^{-1} x$, the y axis, the line $y = \pi$ is rotated one revolution about the y - axis. 3
- c)
 - i) The acceleration in ms^{-2} of an object is given by $\ddot{x} = 2x^3 + 4x$. If the object is initially 2 m to the right of the origin travelling with velocity $8ms^{-1}$, find an expression for v^2 (the square of its velocity) in terms of x . 2
 - ii) What is the minimum speed of the object?
Give a reason for your answer. 2

Question 6 (12 marks)**Marks**

- a) Steven borrows \$50 000 to pay for a new car. He plans to repay the loan by making 60 equal monthly instalments. Interest is charged at the rate of 0.6% per month on the balance owing.
- i) Show that immediately after making two monthly instalments of \$M, the balance owing is given by $\$(50601.80 - 2.006M)$ **2**
- ii) Calculate the value of each monthly instalment. **2**
- b) Prove by Mathematical Induction that:
 $2(1!) + 5(2!) + 10(3!) + \dots + (n^2 + 1)n! = n(n + 1)!$ for positive integers $n \geq 1$. **4**
- c) If $y = \frac{\log_e x}{x}$
- i) Show that $\frac{dy}{dx} = \frac{1 - \log_e x}{x^2}$ **1**
- ii) and hence or otherwise show that
$$\int_e^{e^2} \frac{1 - \log_e x}{x \log_e x} dx = \log_e 2 - 1$$
 3

Question 7 (12 marks)

- a) The deck of a ship was 1.4 m below the level of a wharf at low tide and 0.4 m above wharf level at high tide. Low tide was at 8:24 am and high tide at 2:40 pm. If the tide's motion is simple harmonic, find the first time after 8:24 am that the deck was level with the wharf. **5**
- b) i) Show that $(1+x)^m \left(1 - \frac{1}{x}\right)^m = \left(x - \frac{1}{x}\right)^m$ **2**
- ii) By considering the term(s) independent of x in the expansion of part b (i), justify the result:
$$\binom{2002}{0}^2 - \binom{2002}{1}^2 + \binom{2002}{2}^2 - \dots + \binom{2002}{2002}^2 = -1 \binom{2002}{1001}$$
 5

END OF PAPER

Trial Higher School Certificate Examination – Extension I Mathematics Solutions

Question 1

a)

$$\begin{aligned} D_{\perp} &= \left| \frac{3(1) - 4(4) - 2}{\sqrt{9 + 16}} \right| & 3x - 2 - 4y &= 0 \\ &= \left| \frac{-15}{\sqrt{25}} \right| \\ &= \frac{15}{5} \\ &= 3 \end{aligned}$$

b)

$$\begin{aligned} &2(x^3 - 64) \\ &= 2(x - 4)(x^2 + 4x + 16) \end{aligned}$$

c) i) $y' = 2 \cos 2x$

$$\begin{aligned} \text{ii) } y &= \log_e \left(\frac{2x - 1}{3x + 2} \right)^{\frac{1}{2}} \\ y &= \frac{1}{2} [\log_e (2x - 1) - \log_e (3x + 2)] \\ y' &= \frac{1}{2} \left[\frac{2}{2x - 1} - \frac{3}{3x + 2} \right] \\ &= \frac{7}{2(2x - 1)(3x + 2)} \end{aligned}$$

d) $P(2) = 2^4 - 2(2)^3 - 3 = -3$

e)

$$\begin{aligned} \text{e) } &\int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \frac{dx}{\sqrt{4 - x^2}} \\ &= \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \\ &= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \\ &= \frac{\pi}{3} - \frac{\pi}{4} \\ &= \frac{\pi}{12} \end{aligned}$$

Question 2

a)

$$\tan \theta = \left| \frac{-2-3}{1+(-6)} \right|$$

$$= 1$$

$$\therefore \theta = 45^\circ$$

$$2x + y = 17 \Rightarrow m_1 = -2$$

$$3x - y = 3 \Rightarrow m_2 = 3$$

b)

$$\int_0^3 \frac{y}{\sqrt{y+1}} dy$$

$$= \int_1^2 \frac{u^2 - 1}{u} 2u \cdot du$$

$$= 2 \int_1^2 (u^2 - 1) du$$

$$= 2 \left[\frac{u^3}{3} - u \right]_1^2$$

$$= 2 \left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right)$$

$$= 2 \frac{2}{3}$$

$$y = 0, u = 1$$

$$y = u^2 - 1 \quad y = 3, u = 2$$

$$\frac{dy}{du} = 2u$$

$$\therefore dy = 2u du$$

c)

$$\frac{\cos 2\theta}{\cos \theta - \sin \theta} = \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta}$$

$$= \cos \theta + \sin \theta$$

$$= RHS$$

QED

d)

$$\frac{1}{x(2-x)} \leq 0 \quad x \neq 0, 2$$

$$x(2-x) \leq 0$$

$$x = 0 \quad x = 2 \quad S = \{x : x < 0, x > 2\}$$



Question 3

a) P (-7, 3) Q (9, 15) B (14, 0)

$$\begin{matrix} \times \\ 3:1 \end{matrix}$$

$$A\left(\frac{-7 + 27}{4}, \frac{3 + 45}{4}\right) = A(5, 12)$$

$$M(PQ) = \frac{15 - 3}{9 + 7} = \frac{3}{4}$$

$$M(AB) = \frac{12 - 0}{5 - 14} = \frac{-4}{3}$$

$$M(PQ) M(AB) = \frac{3}{4} \times \frac{-4}{3} = -1$$

$$\therefore PQ \perp AB \text{ (prod of } M = -1)$$

b)

$$\text{Domain } -\frac{1}{3} \leq x \leq \frac{1}{3}$$

$$\text{Range } -\pi \leq y \leq \pi$$

c)

$$T_{r+1} = {}^6C_r (3x)^{6-r} \left(\frac{2}{\sqrt{x}}\right)^r$$

$$= {}^6C_r 3^{6-r} 2^r x^{6-r} x^{-\frac{1}{2}r}$$

$$= {}^6C_r 3^{6-r} 2^r x^{6-\frac{1}{2}r}$$

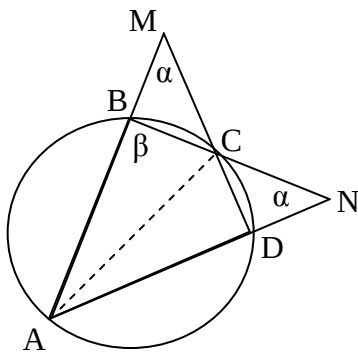
for constant term degree of $x = 0$

$$\therefore 6 - \frac{1}{2}r = 0$$

$$r = 4 \therefore {}^6C_4 3^2 2^4$$

$$= 2160$$

d)



i) Let $\angle AMD = \alpha = \angle ANB$ and $\angle ABN = \beta$

$$\angle BCM = \beta - \alpha \text{ (ext } \angle \text{ of } \Delta)$$

$$\angle DCN = \beta - \alpha \text{ (vert opp)}$$

$$\therefore \angle ADC = \beta \text{ (ext } \angle \Delta)$$

ii) $\angle ABC + \angle ADC = 180^\circ$ (opp angle of cyclic quad suppl)

$$\therefore 2\beta = 180^\circ \therefore \beta = 90^\circ \therefore AC \text{ is diam.}$$

Question 4

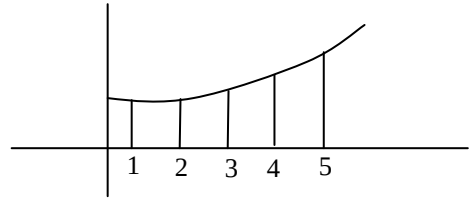
a)

$$A \doteq \frac{b-a}{6} [f(a) + 4\text{odds} + 2\text{evens} + f(b)]$$

$$= \frac{1}{3} [\ln(e+1) + 4\ln(e^2+1) + 2\ln(e^3+1) + 4\ln(e^4+1) + \ln(e^5+1)]$$

$$= \frac{1}{3} [1 + 8.5077 + 6.09717 + 20.02686 + 1.79175]$$

$$= 12.4745$$



iii)

$$f(x) = \ln(e^x + 1)$$

$$f'(x) = \frac{e^x}{e^x + 1}$$

$$f''(x) = \frac{e^x}{(e^x + 1)^2} > 0 \text{ for all } x$$

$\therefore f(x)$ is concave up

$\therefore$ trapezium rule will over estimate area.

x	1	2	3	4	5
$f(x)$	$\ln(e+1)$	$\ln(e^2+1)$	$\ln(e^3+1)$	$\ln(e^4+1)$	$\ln(e^5+1)$

b)

i) $A_1 = 3A_2$

Now $A_2 = \frac{1}{2} r^2 \theta$

$\therefore A_1 = \frac{3}{2} r^2 \theta$

but

$A_1 = A_r - A_2$ where A_r is area of entire region

$$= \frac{1}{2} (OC^2 \theta - r^2 \theta)$$

$$\frac{3}{2} r^2 \theta = \frac{1}{2} \theta (OC^2 - r^2)$$

$$3r^2 = OC^2 - r^2$$

$$4r^2 = OC^2$$

$$\therefore OC = 2r$$

but

$$OC = AC + AO$$

$$2r - r = AC$$

$$\therefore AC = r$$

ii)

$$l_1 = 2r \theta$$

$$l_2 = r \theta$$

$$\therefore \text{tot } l = 2r + 2r + 2r\theta + r\theta$$

$$24 = 4r + 3r\theta$$

$$\therefore \frac{24 - 4r}{3r} = \theta$$

Question 4 continued.

iii)

$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} r^2 \left(\frac{24 - 4r}{3r} \right)$$

$$= \frac{24}{6} r - \frac{2r^2}{3}$$

$$A'' = \frac{-4}{3} \quad \therefore \text{max area, concave down}$$

$$A' = 4 - \frac{4r}{3}$$

$$\therefore \frac{4r}{3} = 4$$

$$r = 3 \text{ gives maximum area}$$

Question 5

a)

$$\cos \theta + 3 \sin \frac{\theta}{2} - 2 = 0$$

$$1 - 2 \sin^2 \frac{\theta}{2} + 3 \sin \frac{\theta}{2} - 2 = 0$$

$$2 \sin^2 \frac{\theta}{2} - 3 \sin \frac{\theta}{2} + 1 = 0$$

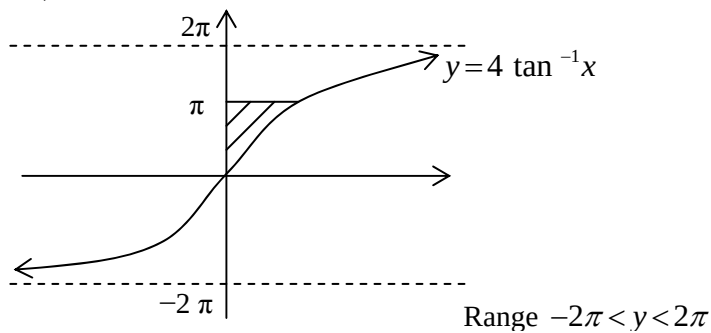
$$\left(2 \sin \frac{\theta}{2} - 1 \right) \left(\sin \frac{\theta}{2} - 1 \right) = 0$$

$$\sin \frac{\theta}{2} = \frac{1}{2}, 1$$

$$\therefore \frac{\theta}{2} = \frac{\pi}{6}, \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{3}, \pi$$

b) i)



Question 5 continued

b) ii)

$$\frac{y}{4} = \tan^{-1} x$$

$$x = \tan \frac{y}{4}$$

$$\begin{aligned} V &= \pi \int_0^{\pi} x^2 dy \\ &= \pi \int_0^{\pi} \tan^2 \frac{y}{4} dy \\ &= \pi \int_0^{\pi} (\sec^2 \frac{y}{4} - 1) dy \\ &= \pi \left[4 \tan \frac{y}{4} - y \right]_0^{\pi} \\ &= \pi \left[\left(4 \tan \frac{\pi}{4} - \pi \right) - (4 \tan 0 - 0) \right] \\ &= \pi (4 - \pi) u^3 \end{aligned}$$

c) i)

$$\ddot{x} = 2x^3 + 4x$$

$$\frac{d\left(\frac{1}{2}v^2\right)}{dx} = 2x^3 + 4x$$

$$\frac{1}{2}v^2 = \frac{x^4}{2} + 2x^2 + c$$

$$t = 0, x = 2, v = 6$$

$$\therefore 18 = 8 + 8 + c$$

$$c = 2$$

$$\therefore v^2 = x^4 + 4x^2 + 4$$

ii)

$$v^2 = (x^2 + 2)^2$$

$$\therefore v^2 \geq 4, \therefore v \neq 0$$

$\therefore$ object never changes direction.

Always moves to right with increasing speed since initial velocity >0 and acceleration >0 for $x > 0 \quad \therefore$ min speed is initial speed.

$\therefore$ min speed is $6ms^{-1}$.

Question 6

- a) i) Amount owing after 1st payment: = $\$50000 \times 1.006 - M$.
 Amount owing after 2nd payment = $\$50000 \times 1.006 - M(1 + 1.006)$
 Balance = $\$50601.80 - 2006 M$

ii)

$$30000 \times 1.006^{60} - \frac{M(1.006^{60} - 1)}{0.006} = 0$$

$$\therefore M = \frac{50000 \times 1.006}{\frac{1.006^{60} - 1}{0.006}}$$

$$\therefore M = \$994.78$$

b)

When $x = 1$, $LHS = 2(1!) = 2$

$$RHS = 1(2!) = 2$$

$\therefore$ true for $n = 1$

assume true for $n = k$

ie $2(1!) + 5(2!) + \dots + (k^2 + 1)k! = k(k + 1)!$

to prove true for $n = k + 1$

ie $2(1!) + 5(2!) + \dots + (k^2 + 1)k! + \left[(k + 1)^2 + 1 \right] (k + 1)! = (k + 1)(k + 2)!$

Now $LHS = 2(1!) + 5(2!) + \dots + (k^2 + 1)k! + (k^2 + 2k + 2)(k + 1)!$

$$= k(k + 1)! + (k^2 + 2k + 2)(k + 1)! \text{ by assumption}$$

$$= (k + 1)! \{k + k^2 + 2k + 2\}$$

$$= (k + 1)! (k^2 + 3k + 2)$$

$$= (k + 1)! (k + 2)(k + 1)$$

$$= (k + 2)! (k + 1)$$

$$= RHS$$

$\therefore$ if true for $n = k$, then true for $n = k + 1$ and since true for $n = 1$,
 then true for all integers, $n \geq 1$

c) i)

$$\begin{aligned} \frac{dy}{dx} &= \frac{x \times \frac{1}{x} - (1)(\ln x)}{x^2} \\ &= \frac{1 - \ln x}{x^2} \end{aligned}$$

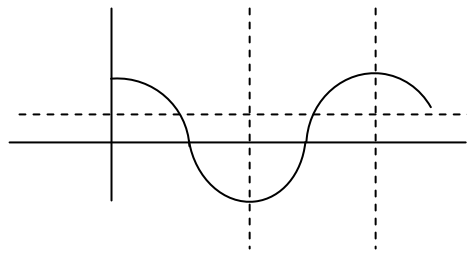
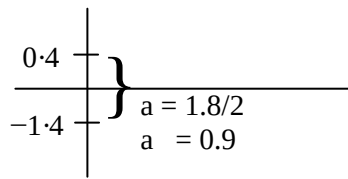
Question 6 continued

c) ii)

$$\begin{aligned}\int_e^{e^2} \frac{1 - \ln x}{x \ln x} dx &= \int_e^{e^2} \frac{\frac{1 - \ln x}{x^2}}{\frac{\ln x}{x}} dx \\&= \left[\ln \left(\frac{\ln x}{x} \right) \right]_e^{e^2} \\&= \ln \left(\frac{\ln e^2}{e^2} \right) - \ln \left(\frac{\ln e}{e} \right) \\&= \ln \left(\frac{2}{e^2} \right) - \ln \left(\frac{1}{e} \right) \\&= \ln \left(\frac{2}{e} \right) \\&= \ln 2 - \ln e \\&= \ln 2 - 1\end{aligned}$$

Question 7

a)



$$\frac{T}{2} = 6 \text{ hrs } 16 \text{ min} = 6 \frac{4}{15}$$

$$\frac{T}{2} = \frac{2\pi}{2n}$$

$$n = \frac{15\pi}{94}$$

equation of motion

$$x = -a \cos nt$$

$$\therefore 0.5 = -0.9 \cos \frac{15\pi}{94} t$$

$$t = \frac{94}{15\pi} \left[2n\pi \pm \cos^{-1} \left(-\frac{5}{9} \right) \right]$$

$$t = 4.08298$$

$$= 4 \text{ hr } 18 \text{ min.}$$

$\therefore$ First time after low tide deck is level with wharf is 12:42 pm.

b) i)

$$\begin{aligned} (1+x)^m \left(1 - \frac{1}{x}\right)^m &= \left[\left(1+x\right) \left(1 - \frac{1}{x}\right) \right]^m \\ &= \left[1 - \frac{1}{x} + x - 1 \right]^m \\ &= \left(x - \frac{1}{x} \right)^m \end{aligned}$$

ii) Letting $m = 2002$

$$\begin{aligned} LHS &= (1+x)^{2002} \left(1 - \frac{1}{x}\right)^{2002} \\ &= \left[\binom{2002}{0} + \binom{2002}{1}x + \dots + \binom{2002}{r}x^r + \binom{2002}{2002}x^{2002} \right] \\ &\quad \times \left[\binom{2002}{0} - \binom{2002}{1}\frac{1}{x} + \dots + (-1)^r \binom{2002}{r}\frac{1}{x^r} + \dots + \binom{2002}{2002}\frac{1}{x^{2002}} \right] \end{aligned}$$

Co eff of x^0 in LHS is

$$\begin{aligned} &\binom{2002}{0} \times \binom{2002}{0} + \binom{2002}{1} \times -\binom{2002}{1} + \dots + (-1)^r \binom{2002}{r}^2 + \dots + \binom{2002}{2002}^2 \\ \text{ie. } &\binom{2002}{0}^2 - \binom{2002}{1}^2 + \binom{2002}{2}^2 - \dots + (-1)^r \binom{2002}{r}^2 + \dots + \binom{2002}{2002}^2 \end{aligned}$$

Question 7 continued

b) ii) $RHS = \left(x - \frac{1}{x}\right)^{2002}$

Gen term is $\binom{2002}{r} x^{2002-r} \left(-\frac{1}{x}\right)^r$
 $= (-1)^r \binom{2002}{r} x^{2002-2r}$

$\therefore$ co eff of x° occurs when $2002 - 2r = 0$

ie. $r = 1001$

$\therefore$ co eff is $(-1)^{1001} \binom{2002}{1001} = -1 \binom{2002}{1001}$

$\therefore \quad \binom{2002}{0}^2 - \binom{2002}{1}^2 + \binom{2002}{2}^2 - \dots + \binom{2002}{2002}^2 = -1 \binom{2002}{1001}$

THE END